

Unit - I Basics of AI and Problem Solving Techniques

1.1 Basic Definition and Terminology of AI

Foundation and Evaluation of AI

What is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science that focuses on creating machines or systems capable of performing tasks that normally require human intelligence.

These tasks include:

- i. Learning ii. Reasoning iii. Problem-solving iv. Decision-making v. Understanding language vi. Recognizing images and speech

Historical Foundation of AI

Early Development

- 01. The AI concept started in the 1950s.
- 02. The term “Artificial Intelligence” was introduced by John McCarthy in 1956.
- 03. AI evolved from math, psychology, neuroscience, & computer science.

Foundation

- 01. **Data & Knowledge Representation:** Storing and organizing information.
- 02. **Algorithms:** Steps that help AI learn and decide.
- 03. **Statistics & Mathematics:** Finding patterns and predictions.
- 04. **Neuroscience & Cognitive Science:** Imitating human thinking.
- 05. **Philosophy & Ethics:** Making AI fair and responsible.

Terminology

- 01. **Agent:** Perceives environment and acts accordingly.
Ex: Robot vacuum cleaner.
- 02. **Machine Learning (ML):** Systems learn from data.
Ex: Email spam detection.
- 03. **Neural Network (NN):** Brain-inspired computing model.
Ex: Cat image recognition.

04. **Deep Learning (DL):** Multilayer neural networks.

Ex: Self-driving cars.

05. **Natural Language Processing (NLP):** Understanding human language.

Ex: Siri and Alexa.

06. **Computer Vision (CV):** Understanding visual data.

Ex: Facebook photo tagging.

07. **Turing Test / Chatbot:** Testing human-like machine intelligence.

Ex: ChatGPT c YBHonversations.

Evaluation of AI

AI systems are evaluated based on how effectively they perform intelligent tasks.

Common Evaluation Criteria

Criteria	Meaning
Accuracy	Correctness of results
Speed	Time taken to complete tasks
Fairness	Ability to treat everyone equally without bias.
Robustness	Ability to handle unusual or difficult situations.

Scope of AI

01. Machine Learning (ML) 02. Natural Language Processing (NLP) 03. Robotics 04. Computer Vision 05. Expert Systems

Components of AI

01. **Data:** Information used for learning.

02. **Algorithms:** Methods for processing and decisions.

03. **Computing Power:** Hardware for AI processing.

04. **ML Models:** Systems that learn from experience.

05. **NLP:** Understanding human language.

06. **Robotics:** Machines working in the real world.

07. **UI:** Allows users to interact with AI systems.

Types of AI

Based on Capabilities

01. **Narrow AI (Weak AI):** Performs a specific task only. **Eg:** Siri, Alexa.
02. **General AI (Strong AI):** Perform human-like intelligence. **Eg:** Space explore.
03. **Super AI:** Surpasses humans. **Example:** Ex Machina or Her.

Based on Functionality

01. **Reactive Machines:** Respond only to current situations. **Eg:** IBM Deep Blue.
02. **Limited Memory AI:** Uses past data to make decisions. **Eg:** Self-driving cars.
03. **Theory of Mind AI:** Understands emotion & thought. **Eg:** humanoid robots.
04. **Self-Aware AI:** Has self-consciousness and awareness. **Eg:** Ultron.

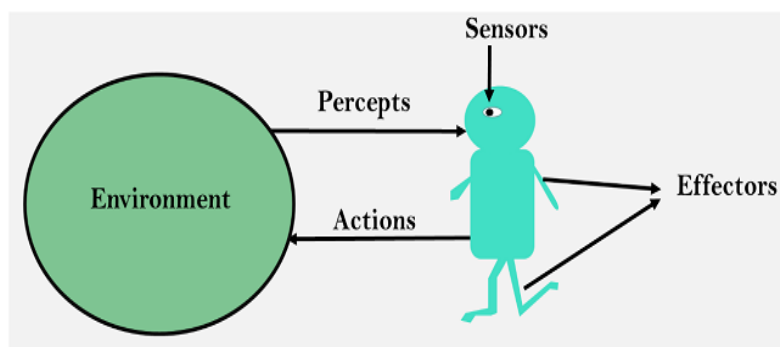
Applications of AI

01. Healthcare 02. Education 03. Banking & Finance 04. Transportation 05. Robotics 06. Entertainment 07. Agriculture 08. Security & Surveillance

1.2 Intelligent Agent in AI

Structure of Agents

An agent is an entity that perceives its environment through sensors and acts upon it using actuators.



Components:

01. **Performance Measure:** How an agent performs. Example: Self-driving car: Speed and reaching destination.
02. **Environment:** World where agent operates. Example: Roads, traffic, signals.
03. **Actuators:** Parts used to take action. Example: Steering, brakes, accelerator.
04. **Sensors:** Parts used to collect information. Example: Cameras, radar, GPS.

Types of AI agent

01.Simple Reflex Agent

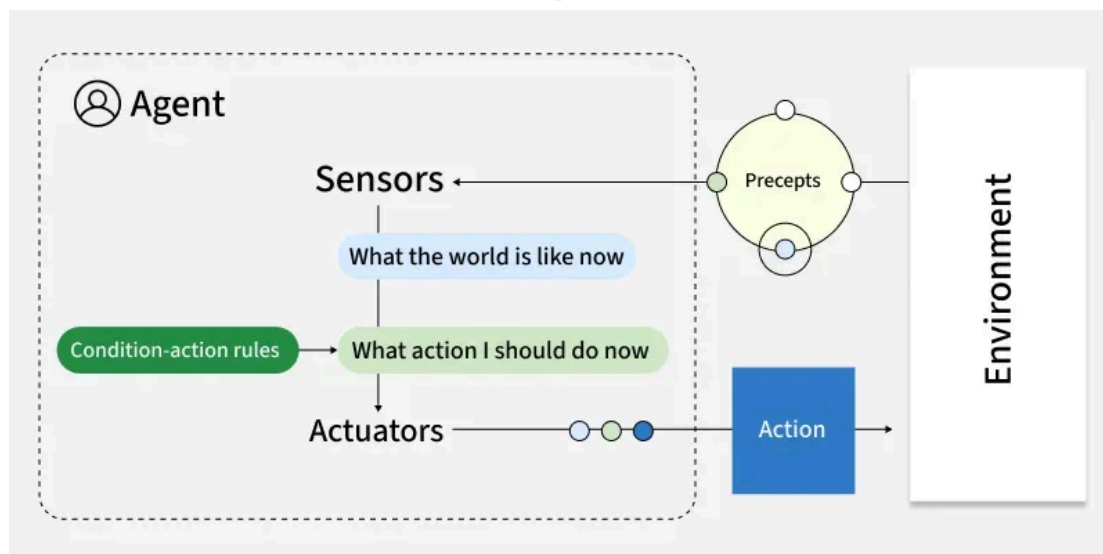
A simple reflex agent in AI is the most basic type of agent that acts only based on the current input (what it senses right now) using fixed if-then rules.

Working:

- A. Sensors read the environment to find "what the world is like now."
- B. Matches rules: It applies fixed Condition-action rules to the current state.
- C. Decides action: It instantly determines "what action I should do now."
- D. Executes output: Actuators deploy the Action back into the environment.
- E. Lacks memory: It operates entirely on the present moment with no history.

Example:

Microwave acts like "timer ends → stop heating"



02. Model-Based Reflex Agent

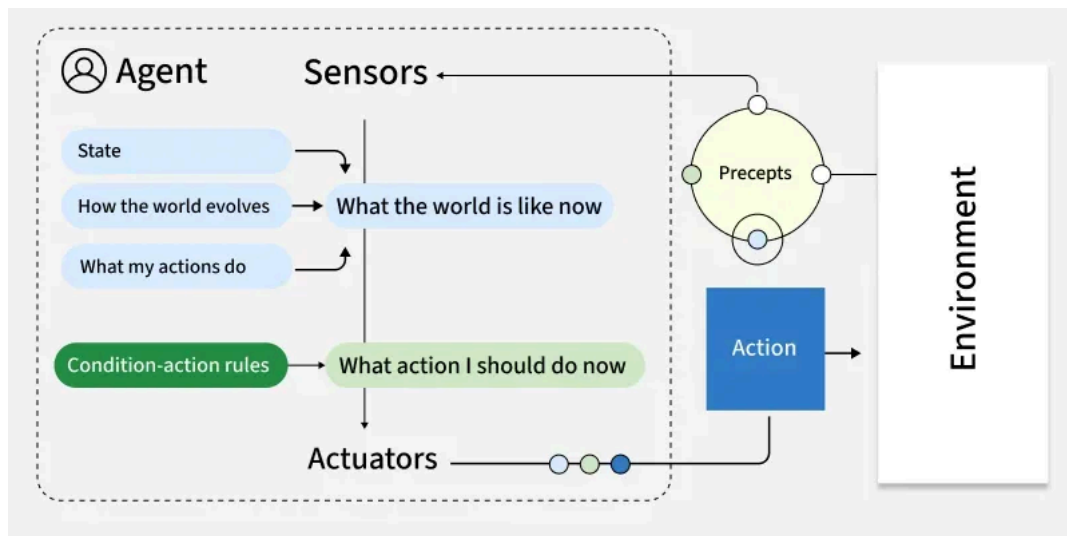
Agent that makes decisions based on the current input and an internal model (memory predefined) of how the world works.

Working:

- A. Combines sensory input with internal State history to see "what the world is like now."
- B. Factor in "how the world evolves" and "what my actions do."
- C. Applies Condition-action rules to this updated internal model.
- D. Instantly determines "what action I should do now."
- E. Actuators deploy the Action back into the environment.

Example:

A robot vacuum remembers cleaned areas and avoids them while cleaning the room.



03. Goal-Based Agent

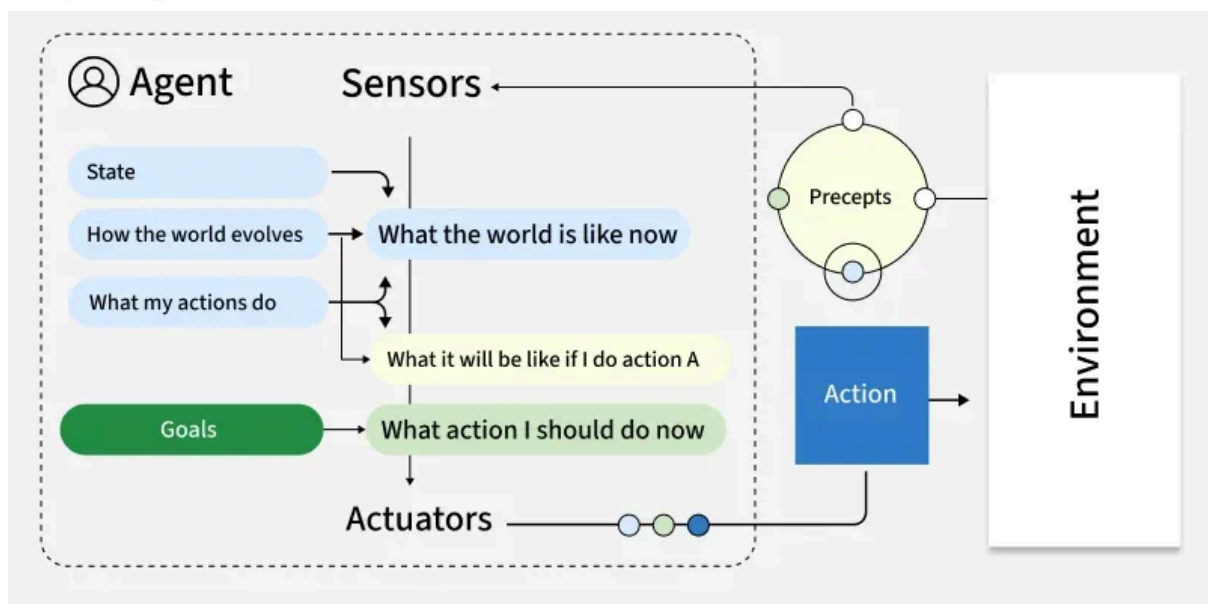
A goal-based agent in AI makes decisions by choosing actions that help it achieve a specific goal, considering future outcomes rather than just current input.

Working:

- Takes input from sensors (current environment)
- Uses internal model of the world
- Evaluates possible future actions
- Selects action that helps achieve the goal
- Produces goal-oriented action

Example:

Google Maps choosing the fastest route to reach a destination.



04. Utility-Based Agent

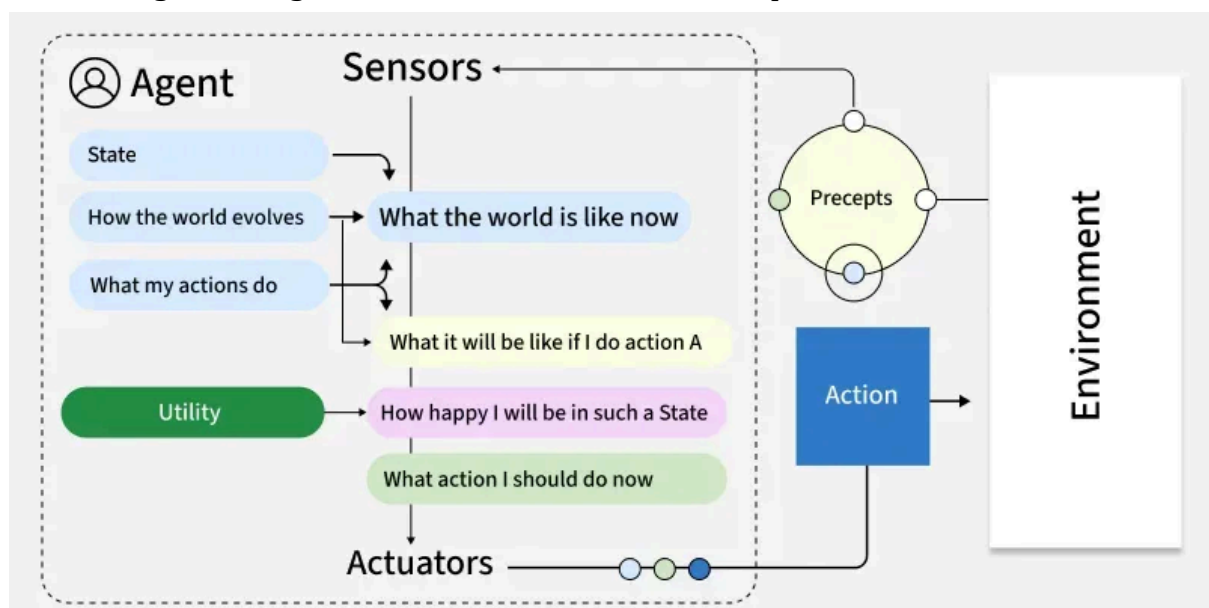
A utility-based agent in AI chooses actions based on which outcome gives the highest “utility” (best satisfaction, efficiency, or benefit), not just reaching a goal.

Working:

- A. Takes input from sensors
- B. Uses internal model of world
- C. Assigns utility (value) to different outcomes
- D. Compares and selects best possible action
- E. Produces most beneficial action

Example:

Self driving choosing the short time and fastest ride option.



05. Learning Agent

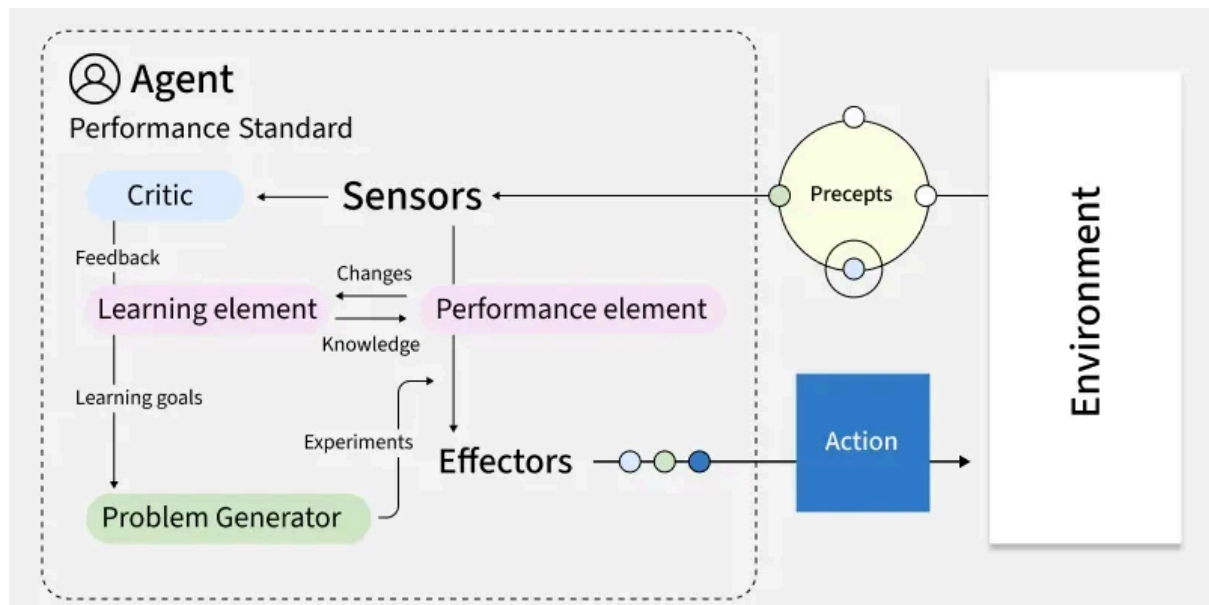
A learning agent in AI improves its performance over time by learning from experience and feedback.

Working:

- A. Takes input from sensors (environment)
- B. Learns from past actions and feedback
- C. Updates knowledge base or model
- D. Improves decision-making over time
- E. Produces better actions with experience

Example:

A spam email filter that gets better at detecting spam over time.



Concept of Rationality

Rationality means: Choosing the best action to achieve a goal.

A rational agent: Observes environment, Takes correct actions, Maximizes performance

Example: Self-driving car choosing safest route.

Nature of Environment

The environment is the surroundings in which an AI agent works.

Types of Environment:

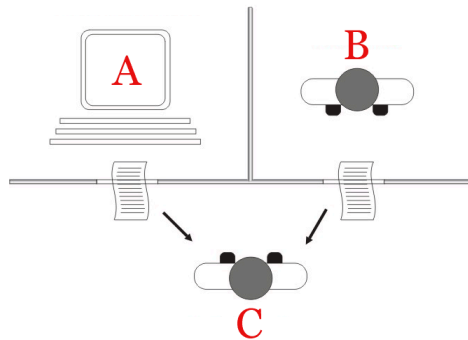
Type	Meaning	Example
Fully Observable	Complete information available	Chess
Partially Observable	Incomplete information	Driving
Deterministic	Predictable outcome	Calculator
Stochastic	Random outcome	Weather
Episodic	Independent tasks	Spam filter
Sequential	Current action affects future	Chess
Static	Environment unchanged	Crossword
Dynamic	Changes continuously	Traffic

Turing Test

Proposed by: Alan Turing

A machine passes the Turing Test if a human cannot distinguish it from another human during conversation.

If a human judge cannot distinguish between: Human responses Machine responses then the machine is considered intelligent.



1.3 Search Algorithms in Artificial Intelligence

Properties of Search Algorithms

01. **Completeness:** The algorithm is complete if it always finds a solution when one exists.
02. **Optimality:** The algorithm is optimal if it finds the best or lowest-cost solution.
03. **Time Complexity:** Amount of time required to find the solution.
04. **Space Complexity:** Amount of memory used during search.

Types of Search Algorithms

01. Uninformed Search Algorithms

These algorithms search without any extra knowledge about the goal.

Types:

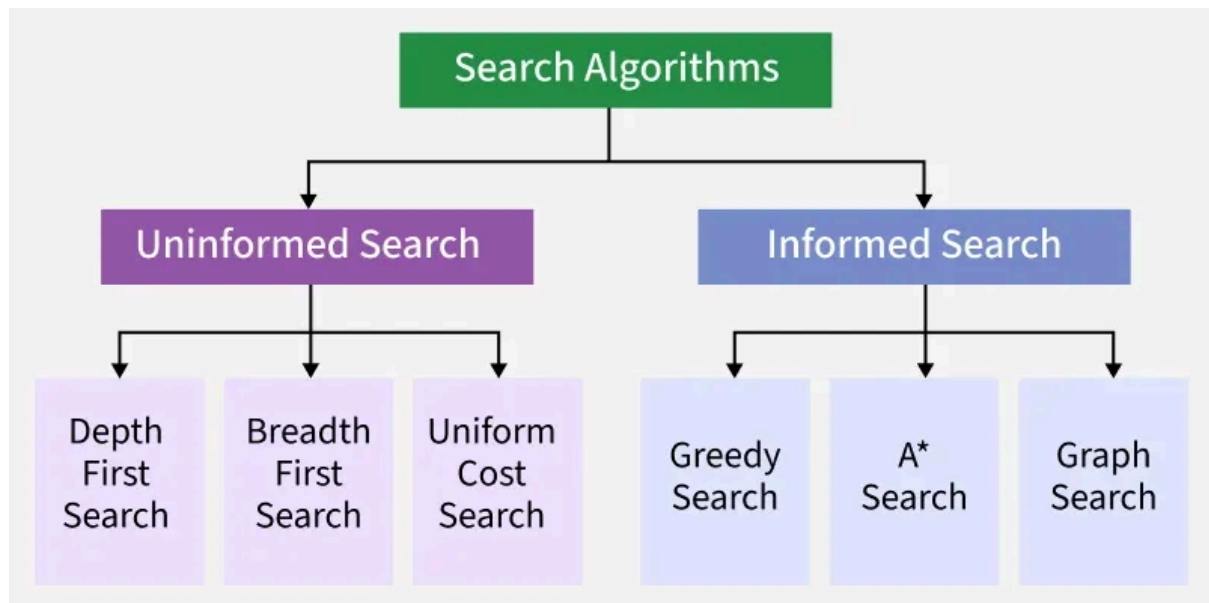
- A. Breadth First Search (BFS)
- B. Depth First Search (DFS)
- C. Uniform Cost Search (UCS)

02. Informed Search Algorithms

Also called Heuristic Search. These algorithms use heuristic information to reach the goal efficiently.

Types:

- A. Greedy Best First Search
- B. A* Search Algorithm
- C. Graph Search



1.4 Heuristic Search Techniques

Heuristic search techniques use special knowledge (heuristics) to solve problems faster and more efficiently.

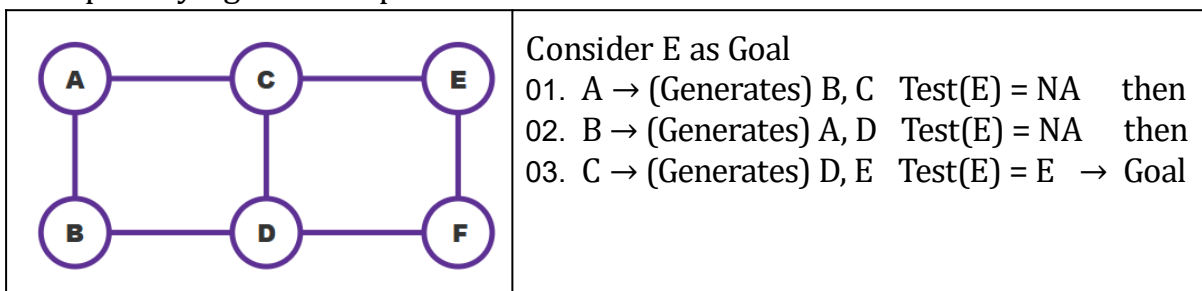
Generate-and-Test

In this method:

1. Generate possible solutions
2. Test whether the solution satisfies the goal

The process continues until a correct solution is found.

Example: Trying different passwords until the correct one is found

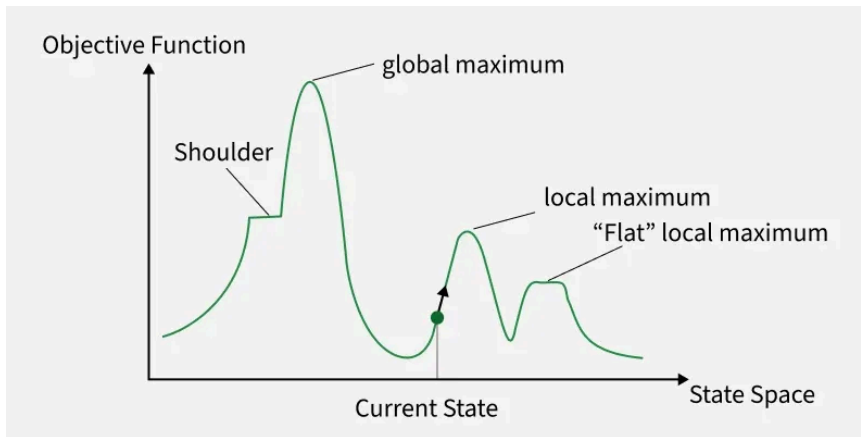


Hill Climbing

Hill Climbing is an optimization technique that moves step-by-step toward a better state. It selects the best neighboring state until the goal is reached.

Example: Finding the shortest route on a map.

Limitation May get stuck at local optimum.

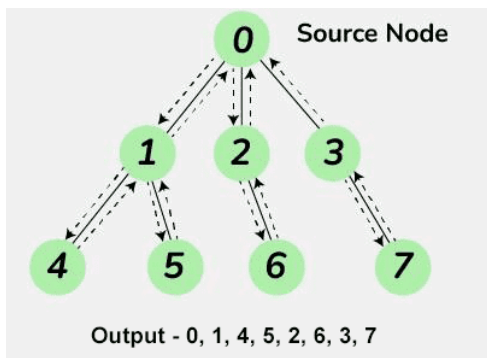


Depth-First Search (DFS)

DFS explores the deepest node first before backtracking.

- Uses Stack (LIFO)
- Requires less memory

Example Maze solving

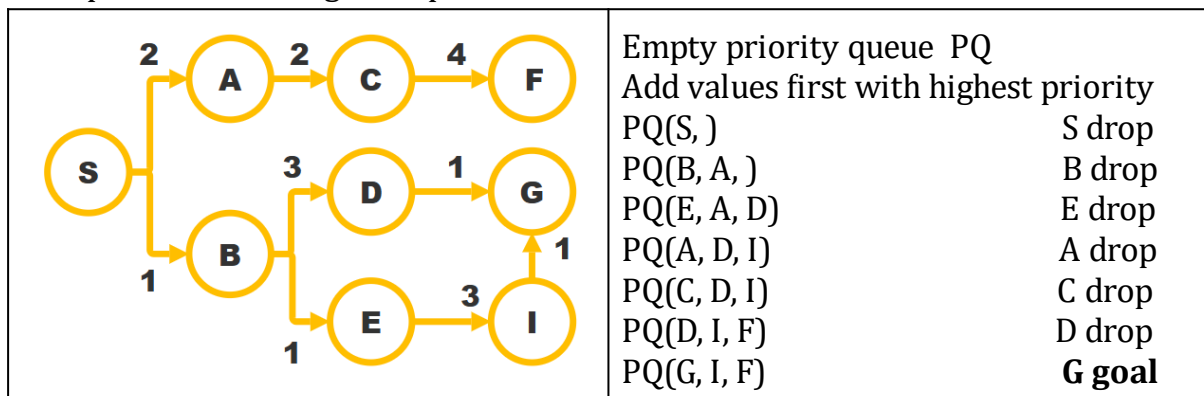


Best-First Search

Best-First Search selects the most promising node using heuristic values.

- Faster than blind search
- Uses evaluation function

Example: Route finding in maps



Greedy Best-First Search

This search chooses the node that appears closest to the goal.

Uses heuristic function:

$$f(n)=h(n)$$

A* Algorithm

A* combines: Actual path cost and Heuristic estimated cost

Formula: $f(n)=g(n)+h(n)$

Where:

- $g(n)$ = actual cost
- $h(n)$ = estimated cost

Example: GPS navigation systems

Problem Reduction

A complex problem is divided into smaller sub-problems for easier solving.

Example Breaking a large project into smaller tasks

Used in:

- Expert systems
- Planning problems

Used in: Expert systems, Planning problems

1.5 Beyond Classical Search

Beyond classical search methods are advanced techniques used when problems are complex, dynamic, or uncertain.

Local search algorithms and optimization problem

Local search algorithms improve a solution step-by-step by moving to a better neighboring state.

Example: Traveling Salesman Problem (TSP)

Local search in continuous spaces

In continuous spaces, states can take infinite values instead of fixed discrete values.

Example: Robot movement

Searching with nondeterministic action and partial observation

Sometimes actions may produce unpredictable results, and the agent may not have complete information.

Example: Self-driving cars in bad weather

Online search agent and unknown environments

An online search agent learns and searches while interacting with the environment.

Example: Robot exploring a new building without a map